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Xiaolin Sun and Yanhua Wang*

Mathematics and Statistics, Beijing Institute of Technology, Beijing, 100081, China

*Corresponding author's e-mail: yhwang@bit.edu.cn

Abstract: In this paper, we introduce a prediction algorithm founded on conformal prediction, tailored for constructing prediction intervals in the context of censored survival data. Conformal prediction frameworks distinguish themselves from other prediction paradigms by their non-empirical evaluation, reliance on user-defined confidence intervals for modeling errors, and widespread adoption across regression and classification methodologies, inclusive of survival analysis, in recent years. Herein, we present a novel application wherein the Kaplan-Meier method is employed to compute empirical quantiles of nonconformal scores, specifically tailored for censored schematic variables. This novel approach facilitates the generation of well-calibrated prediction intervals for survival times, augmenting any existing survival prediction algorithm. Validation of its efficacy and computational efficiency is performed on both the real-world dataset 'SUPPORT' and the synthetic dataset 'RRNLNPH.'

1. Introduction

Survival analysis, a method widely employed in medical research, serves as a pivotal tool for examining the time until the incidence of a particular event. Such events encompass mortality or disease recurrence in medical contexts, termination of employment in corporate settings, or re-arrest of parolees[1]. Initially applied to assess the longevity of weaponry, survival analysis primarily involves estimating the probability of an entity remaining operational after a specified period, such as a year of storage for artillery shells. This analytical framework aims to forecast the survival function and duration for new subjects. It is noteworthy that the acquisition of longevity data varies, ranging from deliberately designed experiments to field investigations. Consequently, these datasets frequently exhibit characteristics of censorship or imprecision. In both engineering and medical contexts, it is common to predefine a termination point for tests or observations[1][2]. However, until this designated time elapses, certain individuals may neither experience the event nor reach the end of their lifespan. In such instances, the only information available pertains to the survival duration exceeding the predetermined cutoff point. This scenario characterizes right-censored data, wherein the precise survival time remains unknown.

In numerous empirical datasets, the prevalence of censored data is notable, constituting a substantial proportion of all observations. Despite its inherent imprecision, the removal of censored data directly during predictive model generation is ill-advised, as it may introduce significant bias into model outputs. Consequently, various regression modeling strategies have been employed in the context of survival analysis to tackle this issue. Semi-parametric regression techniques, exemplified by the Cox regression model, have garnered attention. Beyond the conventional Cox regression model, a spectrum of extended regularized Cox regression models has emerged, including but not limited to Lasso Cox, Ridge Cox, and Elastic-Cox models[3]. Parametric regression approaches, such as exponential and Weibull models, have also been explored in this domain. Furthermore, the proliferation of medical technology has led to the accumulation of increasingly vast clinical datasets. Concurrently, the field has witnessed rapid advancements in machine learning methodologies tailored for processing and analyzing high-dimensional data[4]. Examples include Random Survival Forest (RSF)[5], Deepsur[6], and DeepHit [7], among others. These models offer the advantage of obviating the need for stringent modeling assumptions[8]. However, they are beset by a notable drawback: the lack of robust quantification of prediction uncertainty. It is our contention that an effective model not only furnishes reasonably accurate outcome predictions but also confers a measure of confidence in these prognostications. Hence, the integration of uncertainty quantification tools becomes imperative in this regard.

One particular model is a machine learning algorithm model, Conformal Prediction[9][10][11], which can be used to produce interval predictions with valid edge coverage without making any distributional assumptions. This algorithm divides the data into a training set and a test set, fits the training set data with a specific algorithm, and calibrates the fitted model using a calibration set, where the concept of inconsistency scores is introduced, and the inconsistency scores are used to perform the calibration to obtain prediction intervals, rather than point predictions[10][12][13]. We want to combine the conformal prediction algorithm to generate prediction intervals for the survival time of deletions with guaranteed coverage, so how to take the deletion of schematic variables into account is what we want to focus on.

On this basis, an idea is given to us: the survival analysis prediction algorithm is fitted on the training set, and then the errors of the fitted model are computed on the calibration set (each error is accompanied by a censored indicative variable). When obtaining the empirical quantile of the error, in order to take into account the censored indicative variable, we combine the Kaplan-Meier algorithm[14] to obtain the distribution of the error first. Then, we calculate the empirical quantile quartiles to calibrate the predicted values to obtain prediction intervals.

2. Preliminaries

2.1. Conformal prediction

Among the plethora of algorithms employed for the quantification of uncertainty, conformal prediction has garnered widespread adoption and application across various domains owing to its straightforward framework and minimal computational overhead. Conceptually, conformal prediction initially partitions the dataset into a training set and a calibration set, applies a machine learning algorithm to the calibration set, and subsequently establishes confidence intervals through the utilization of non-conformity scores on the calibration set[12].

We assume that $D = \{(X_i, Y_i)\}_{i \in I}$ is the dataset, where i is the set of data indices and the samples (X_i, Y_i) obey the distribution P. Typically, covariate prediction only requires that the data satisfy exchangeability (see the assumption below), rather than the stronger condition of being independently and identically distributed[12][13]. First, we have to randomly split the dataset into a training set $D_{tr} = \{(X_i, Y_i)\}_{i \in I_{tr}}$ and a calibration set $D_{cal} = \{(X_i, Y_i)\}_{i \in I_{cal}}$, with the test point (X, Y) being new data sample, and then we fit a model $\hat{\mu}(X)$ given an algorithm $\mathcal{A}\big((X_i, Y_i): i \in I_{tr}\big)$ to train on the training set. Immediately after that, unlike other prediction models, conformal prediction requires a calibration process. Specifically, the trained model $\hat{\mu}(X)$ is used to compute the predicted values $\{\hat{\mu}(X_i): i \in I_{cal}\}$ on the calibration set, and then the error of the model (i.e., the non-consistency score) is computed by comparing it with the true values on the calibration set:$R_i = |Y_i - \hat{\mu}(X_i)|, i \in I_{cal}$. Based on the non-consistency score, the calibration is performed, and thus confidence intervals are constructed. In summary, our task is that given a confidence level and a test point, we need to construct a confidence interval such that the following equation holds.

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Algorithm 1 Conformal Prediction

Input:

Data $D = \{(X_i, Y_i)\}_{i \in I}$, test point (X, Y), confidence level $\alpha \in (0,1)$, regression algorithm $\mathcal{A}$.

Process:

Randomly split data set D into two sets: training set $D_{tr} = \{(X_i, Y_i)\}_{i \in I_{tr}}$ and calibration set $D_{cal} = \{(X_i, Y_i)\}_{i \in I_{cal}}$.

Fit on the training data set: $\hat{\mu}(X) \leftarrow \mathcal{A}\big((X_i, Y_i): i \in I_{tr}\big)$.

Calculate absolute residuals on calibration data set: $R_i = |Y_i - \hat{\mu}(X_i)|, i \in I_{cal}$

For given level α, compute a quantile of the empirical distribution of residuals:

$Q_{1-\alpha}(R, I_{cal}) \coloneqq (1-\alpha)\left(1 + \frac{1}{|I_{cal}|}\right) - th$ the empirical distribution of $\{R_i: i \in I_{cal}\}$

Output:

Prediction interval $C_{1-\alpha}(X) = [\hat{\mu}(X) - Q_{1-\alpha}(R, I_{cal}), \hat{\mu}(X) + Q_{1-\alpha}(R, I_{cal})]$

Assumption 1 (exchangeability[7]). Assume that the data set $\{(X_i, Y_i)\}_{i \in I}$ and the test point (X, Y) are exchangeable. Formally, define Z_i, $i = 1,2, \dots, |I| + 1$ as the above data pair, then Z_i are exchangeable if arbitrary permutation leads to the same distribution, i.e.,

$$\big(Z_1, Z_2, \dots, Z_{|I|+1}\big) \stackrel{d}{=} \big(Z_{\pi(1)}, Z_{\pi(2)}, \dots, Z_{\pi(|I|+1)}\big),$$

with arbitrary permutation π over $\{1,2, \dots, |I| + 1\}$.

Theorem 1([11]) Assume that the data set $\{(X_i, Y_i)\}_{i \in I}$ are exchangeable. For any $\alpha \in (0,1)$, define the conformal band by

$$C_{1-\alpha}(X) = [\hat{\mu}(X) - Q_{1-\alpha}(R, I_{cal}), \hat{\mu}(X) + Q_{1-\alpha}(R, I_{cal})],$$

Then $C_{1-\alpha}(X)$ satisfies

$$P\{Y \in C_{1-\alpha}(X)\} \geq 1 - \alpha.$$

2.2. Kaplan-Meier estimator

The Kaplan-Meier estimator is a nonparametric estimator proposed by Kaplan and Meier in 1958. The Kaplan-Meier estimator can be used to estimate the unknown overall general survival function when the lifetime data are incomplete, i.e., censored. This estimator is later referred to as the product-limited estimator (PL) and is used in many fields like economics[15].

We consider a set of survival data $(X_i, T_i, C_i), i \in I$, where T_i is the survival time and C_i is the censoring time. The observation time is $t_i = min(T_i, C_i)$, i.e., the observed data are $(X_i, t_i, \delta_i), i \in I$, where $\delta_i = \begin{cases} 1, & if\ T_i \leq C_i\ (uncensored) \\ 0, & if\ T_i > C_i\quad (censored) \end{cases}$ is the censored indicative variable. For all for-event survival time data in the sample, a series of interval intervals are created: $t_{(1)} < t_{(2)} < \cdots < t_{(|I|)}$,and we agree that the number of people who are at risk (e.g., have not died) prior to timepoint t_i is n_i, and the number of people who have had an event (e.g., who have died at this timepoint) at timepoint t_i is d_i. The Kaplan–Meier estimator [14]is then given by

$$S_{KM}(t) = \prod_{i=1}^{j} \frac{n_j - d_j}{n_j}, t_{(j)} < t < t_{(j+1)}.$$

We will consider the event probabilities, not the survival probabilities, and for the survival function of the KM estimator output, we can easily obtain its probability distribution function, i.e.,

$$F(t) = 1 - S_{KM}(t).$$

3. Conformal Kaplan-Meier prediction

3.1. The prediction task

When we apply covariate prediction to survival data, the most obvious difference from the traditional use for regression and classification is that each observation time has a censored schematic variable, so we need to choose algorithms that can handle the type of censored data when fitting the model on the training set. There is also the fact that each non-consistency score we get on the calibration set also has a corresponding censored indicative variable, i.e., we want to get the $1-\alpha$ empirical quantile from the data set $\{(R_i, \delta_i), i\epsilon I_{cal}\}$. We thus choose the Kaplan-Meier estimator, a method that can be adapted to censored data, to obtain the corresponding empirical quantile and thus calibrate our prediction intervals.

3.2. Algorithm

Based on the discussions above, we conclude our algorithm in Algorithm 2.

Algorithm 2 Conformal Kaplan-Meier Prediction

Input:

Data $Z = \{(X_i, t_i, \delta_i)\}_{i\in I}$, test point X, confidence level $\alpha \in (0,1)$, survival analysis regression algorithm $\mathcal{A}$.

Process:

Randomly split data set Z into two sets: training set $Z_{tr} = \{(X_i, t_i, \delta_i)\}_{i\in I_{tr}}$ and calibration set $Z_{cal} = \{(X_i, t_i, \delta_i)\}_{i\in I_{cal}}$.

Fit on the training data set: $\hat{\mu}(X) \leftarrow \mathcal{A}\big((X_i, t_i, \delta_i): i \in I_{tr}\big)$.

Calculate absolute residuals on calibration data set: $R_i = |t_i - \hat{\mu}(X_i)|, i \in I_{cal}$ and we get the nonconformal scores data set $\{(R_i, \delta_i), i \in I_{cal}\}$.

Getting the cumulative distribution function using Kaplan-Meier estimator $F(R) = 1 - S_{KM}(R)$.

For given level α, compute a quantile of the empirical distribution of residuals:

$$Q_{1-\alpha}\big((R_i, \delta_i), I_{cal}\big) := F^{-1}(1-\alpha).$$

Output:

Prediction interval $C_{1-\alpha}(X) = \big[\hat{\mu}(X) - Q_{1-\alpha}\big((R_i, \delta_i), I_{cal}\big), \hat{\mu}(X) + Q_{1-\alpha}\big((R_i, \delta_i), I_{cal}\big)\big]$.

4. Experiments

In this section, we present the experimental results of applying Conformal Kaplan-Meier prediction on two datasets, one from synthetic RRNLNPH (from Kvamme et al. (2019)[16]) and the other from real-world data 'SUPPORT' [17].

4.1. Experimental setup

In the pursuit of substantiating the efficacy of both synthetic datasets and real-world data under consideration, a comprehensive array of experiments was meticulously executed. These experiments aimed to validate the empirical findings through rigorous comparison among various algorithms, encompassing both baseline methodologies and our novel algorithm. Additionally, conventional conformal prediction algorithms, unaided by the Kaplan-Meier predictor, were employed as benchmarks to ascertain the effectiveness of our enhancements. For each dataset, across multiple iterations, a systematic protocol was followed. Initially, 100 sample points were randomly designated as test points. Subsequently, eighty percent of the remaining data were randomly allocated as training data, with the residual subset serving as calibration data. This meticulous process ensured the robustness and reliability of the experimental setup. Quantitative assessments were conducted at diverse confidence levels for each algorithm, with a meticulous endeavor to maintain consistency. Specifically, 10 experimental runs were meticulously executed for each confidence level, meticulously ensuring a robust and comprehensive evaluation framework.

For the algorithms, we chose linear Cox regression as the baseline. In addition, we performed CoxPH (Katzman et al., 2018[6]) and CoxCC (Kvamme et al., 2019[16]) (both belonging to Cox-MLP) to return confidence bands using $S(t)$, although they do not contain theoretical guarantees. We integrated our proposed algorithm CKMP with Coxreg, CoxPH, and CoxCC, labeled CoxPH/CoxCC + CKMP. Furthermore, to ensure that the improvement in CKMP with K-M estimation is useful, we tested conventional conformal prediction algorithms that are not improved with the Kaplan-Meier predictor, labeled CoxPH/CoxCC + CP.

For coverage, since the synthetic dataset contains the true survival time column 'duration_true,' i.e., true T_i (censored or not), we define the empirical coverage to measure the accuracy of the algorithm:

$$EC = \frac{1}{|\mathrm{I}_{test}|} \sum_{i \in \mathrm{I}_{test}} \mathbb{I}\left(T_i \in C_{1-\alpha}(X_i)\right) = \frac{1}{100} \sum_{i \in \mathrm{I}_{test}} \mathbb{I}\left(T_i \in C_{1-\alpha}(X_i)\right).$$

As for the real-world data SUPPORT, we first filtered out the uncensored data in the test set:

$$I'_{test} = \left\{(X_i, t_i, \delta_i)_{i \in I_{test}} : \delta_i = 1\right\},$$

We then define the absolute empirical coverage AEC:

$$AEC = \frac{1}{|I'_{test}|} \sum_{i \in I'_{test}} \mathbb{I}\left(T_i \in C_{1-\alpha}(X_i)\right).$$

4.2. Analysis

Analysis of Table 1. We summarise the experimental results at confidence levels of 0.6, 0.8, and 0.95 for RRNLNPH and SUPPORT, respectively, in Table 1. It can be noted that our proposed CKMP algorithm can almost always return prediction intervals with corresponding accuracy according to a specific confidence level, no matter which baseline algorithm is combined with it. It can be noticed more clearly, especially for the synthetic data RRNLNPH, that, compared with the baseline algorithms Coxreg, Coxcc, and Coxph, the conformal prediction framework combined with the baseline algorithms can provide results that are more in line with our needs, indicating that it can return result intervals with similar accuracy to the confidence level we set. The improved framework CKMP combined with the KM estimator can return intervals with higher accuracy and smaller variance of results per experiment compared to the traditional CP (conformal prediction). In other words, the stability and accuracy of the results returned by the baseline algorithm, the traditional CP framework, and the improved CKMP algorithm framework are sequentially progressive.

Average coverage. Figure 1 shows the coverage of the prediction intervals returned by the nine algorithms at different confidence levels under the synthetic dataset RRNLNPH, where we would expect a good algorithm to return higher coverage. We can see from the figure that the conformal prediction framework combined with the survival analysis prediction algorithm returns a higher coverage of prediction intervals, and this is in the context of its definition of coverage of prediction intervals being stricter than the baseline algorithm's definition of coverage. In particular, the baseline algorithms are unstable after the confidence level is greater than 0.8, and the coverage is far below the confidence level, in contrast to the traditional conformal prediction framework and the improved framework, which both return prediction intervals with the desired coverage.

Improvement of rationality. Figure 2 shows the comparison between before and after improvement with KM estimation on the real-world dataset SUPPORT. From the data in Table 1 and Figure 2, it can be clearly seen that compared with the traditional conformal algorithm CP, the coverage of our CKMP algorithm after improvement with K-M estimation is closer to the confidence level we set and has a smaller variance. These features can fully illustrate that the improved version of KM is superior to the traditional framework, which also shows the importance of improving KM estimation when the data has censoring.

Average interval length. Figure 3 and Figure 4 show the interval lengths returned by all the algorithms on two different datasets, where the improved algorithm CKMP is more conservative than the traditional conformal prediction framework, returning intervals of larger average lengths, and further pointing out that CKMP can return prediction intervals with interval lengths that are within a reasonable range and at the same time with higher coverage. If there is a need to return less conservative prediction intervals, we can improve on the weighted version of the conformal prediction framework, i.e., conformal prediction under covariate shift with KM estimation, which can increase the coverage while reducing the interval length.

Table 1. Coverage and variance results for all algorithms at different alpha.

Approach \Conf.	Accuracy on RRNLNPH			Accuracy on SUPPORT		
	0.6	0.8	0.95	0.6	0.8	0.95
Coxreg	0.620±0.011	0.777±0.011	0.848±0.006	0.695±0.009	0.972±0.003	0.998±0.001

Coxcc	0.581±0.012	0.768±0.009	0.849±0.008	0.688±0.009	0.977±0.010	1.000±0.000
Coxph	0.558±0.016	0.754±0.010	0.848±0.007	0.688±0.009	0.979±0.007	1.000±0.000
CP+Coxreg(un-KM)	0.587±0.052	0.789±0.042	0.937±0.025	0.589±0.067	0.787±0.033	0.942±0.022
CP+Coxcc(un-KM)	0.584±0.040	0.818±0.041	0.951±0.022	0.581±0.043	0.794±0.017	0.950±0.026
CP+Coxph(un-KM)	0.590±0.048	0.812±0.029	0.946±0.019	0.758±0.050	0.900±0.025	0.996±0.009
CKMP+Coxreg	0.583±0.044	0.804±0.033	0.964±0.013	0.723±0.043	0.914±0.031	0.986±0.015
CKMP+Coxcc	0.597±0.059	0.803±0.028	0.955±0.013	0.887±0.035	0.962±0.014	0.999±0.001
CKMP+Coxph	0.629±0.036	0.798±0.038	0.952±0.014	0.871±0.060	0.945±0.141	0.962±0.014

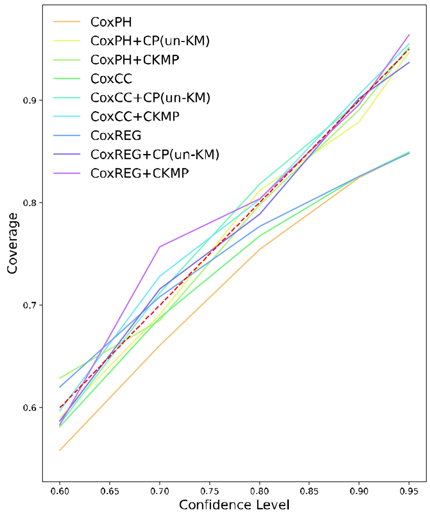

Figure 1. Coverage comparison under different confidence level $(1-\alpha)$ on the synthetic data RRNLNPH.

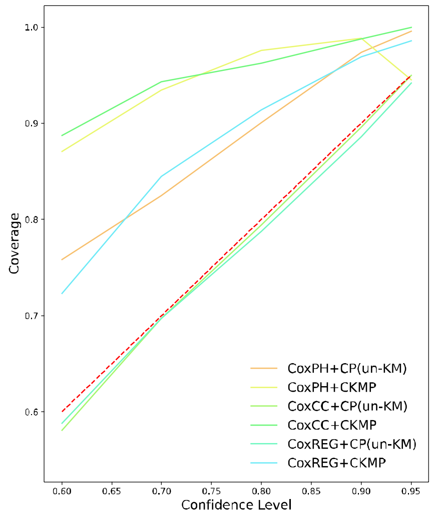

Figure 2. Improving using KM estimator rationality under different confidence level $(1-\alpha)$ on the real-world data SUPPORT.

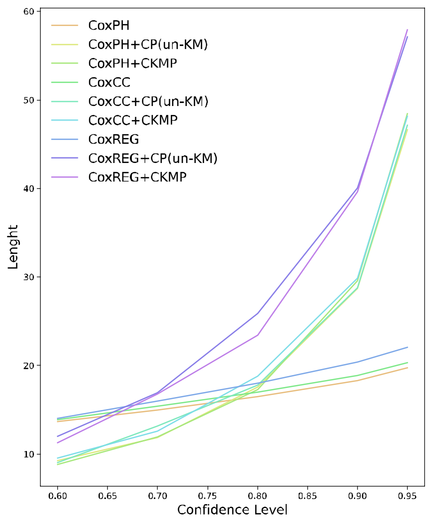

Figure 3. Length comparison under different confidence level $(1-\alpha)$ on the synthetic data RRNLNPH.

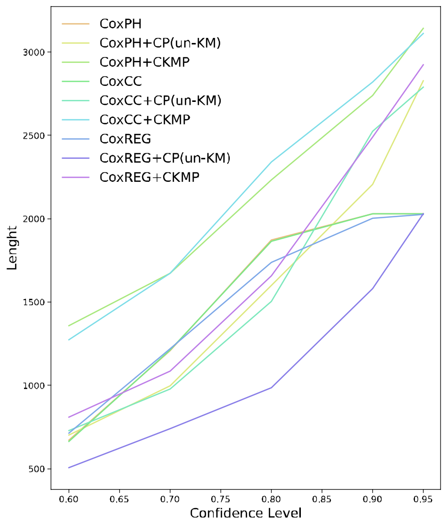

Figure 4. Length comparison under different confidence level $(1-\alpha)$ on the real-world data SUPPORT.

5. Conclusion

In this paper, we propose a conformal prediction framework for censored data based on the Cox model and confidence level to return prediction intervals of survival time. Firstly, we adopt the conformal prediction framework to predict the survival data to get the interval prediction. In order to consider censored, we discard the traditional empirical quartiles to calibrate the prediction intervals, and we choose to improve the traditional KM estimator with the KM estimator. The conformal prediction framework, which in turn can take into account the censored indicative variables in the survival data, is combined with the non-consistency scores to obtain empirical quartiles to calibrate the predicted survival intervals of the test data. Secondly, we conducted a large number of experiments to verify the perfect coverage and small variance of the improved algorithm on both synthetic and real-world data to ensure the upper and lower bounds of the algorithm's guarantees and accuracy.

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